

Session 7 & 8: Python Dictionaries, Files & Lists

Detailed Theory

1. Introducing Dictionaries

- Dictionaries are Python's built-in data structures for storing **key-value pairs**.
 - They are **unordered**, **mutable**, and **indexed by keys** (which can be almost any immutable type).
 - Keys must be **unique** in a dictionary.
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2. Defining Dictionaries

- Dictionaries are defined using curly braces {} with colon : separating keys and values.

- Example:

python

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```
student = {"name": "Komal", "age": 25, "city": "Nagpur"}
```

- Empty dictionary:

python

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```
empty_dict = {}
```

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Using dict() constructor:

```
python
```

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```
d = dict(name="Komal", age=25)
```

3. Modifying Dictionaries

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You can add or update key-value pairs by assigning values to keys:

```
python
```

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```
student["age"] = 26      # update existing  student["course"] = "CS"  # add new key-value
```

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Use update() method to merge dictionaries:

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```
student.update({"city": "Pune", "grade": "A"})
```

4. Deleting Items from Dictionaries

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Use del keyword to remove a key:

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```
del student["grade"]
```

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Use pop() to remove a key and get its value:

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```
age = student.pop("age")
```

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Use `clear()` to empty the dictionary:

python

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`student.clear()`

5. Files in Python

- Python uses built-in functions to **work with files**: open, read, write, and close.
- `open()` opens a file; mode parameter controls behavior ('r' for read, 'w' for write, 'a' for append).
- Example:

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```
f = open("file.txt", "r") content = f.read() f.close()
```

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Using with statement automatically closes file:

python

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```
with open("file.txt", "w") as f: f.write("Hello World")
```

6. Introducing Lists

- Lists are **ordered, mutable** collections of elements.
 - Elements can be of **different types**.
 - Lists are defined using square brackets [].
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7. Defining Lists

- Example:

```
python
```

```
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```

```
numbers = [1, 2, 3, 4] names = ["Komal", "Ankit", "Neha"] mixed = [1, "Hello", 3.14]
```

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Empty list:

```
python
```

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```
empty_list = []
```

8. Adding Elements to Lists

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Use `append()` to add single element at the end:

python

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numbers.append(5)

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Use insert(index, element) to add at a specific position:

python

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numbers.insert(2, 10)

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Use extend() to add elements from another list:

python

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```
numbers.extend([6, 7])
```

9. Searching Lists

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Use in keyword to check if element exists:

python

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```
if 3 in numbers: print("Found")
```

- Use `index()` to get first index of element (raises error if not found):

python

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```
pos = numbers.index(2)
```

10. Deleting List Elements

- Use `del` to remove by index:

python

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```
del numbers[2]
```

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Use `remove(value)` to delete first matching element:

```
python
```

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```
numbers.remove(4)
```

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Use `pop()` to remove and return last (or specified) element:

```
python
```

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```
last = numbers.pop()
```

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Use `clear()` to empty the list:

```
python
```

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```
numbers.clear()
```

11. Using List Operators

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`+` operator concatenates two lists:

python

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```
[1, 2] + [3, 4] # [1, 2, 3, 4]
```

- * operator repeats the list:

python

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```
[1] * 3 # [1, 1, 1]
```

- in checks for membership.
- Lists support slicing to get sublists:

python

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numbers[1:4]